

COIN CHANGE PROBLEM 6

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Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1 .

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers V and N . V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1 .

Sample Input 0

```
7 2
2 4
```

Sample Output 0

```
-1
```

Sample Input 1

```
11 4
1 5 6 9
```

Sample Output 1

```
2
```

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Language

Python 3



```
1 V, N = map(int, input().split())
```

```

2 coins = list(map(int, input().split()))
3 def solve(V):
4     if V == 0:
5         return []
6     best = None
7     for coin in coins:
8         if coin <= V:
9             result = solve(V - coin)
10            if result is not None:
11                result = result + [coin]
12                if best is None or len(result) < len(best):
13                    best = result
14    return best
15 best = solve(V)
16 if best is None:
17     print(-1)
18 else:
19     print(len(best))
20

```

Line: 20 Col: 1



Upload Code as File

☐

Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

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1	7 2
2	2 4

Your Output (stdout)

1	-1
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Expected Output

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1	-1
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